

A High Order Finite Volume Method for Elliptic Interface Problems

Will Thacher^{1,2}, Hans Johansen¹, Dan Martin¹

¹Lawrence Berkeley National Lab

²UC Berkeley

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Problem Statement

Given a two-dimensional domain Ω separated into subdomains $\Omega^\pm$ by the interface Γ , we want to solve:

$$\alpha u - \nabla \cdot (\beta \nabla u) = f \text{ on } \Omega \quad (1)$$

$$[u] = w \text{ on } \Gamma \quad (2)$$

$$[\beta \partial_{\mathbf{n}} u] = v \text{ on } \Gamma. \quad (3)$$

with some boundary conditions on $\partial\Omega$. α, β, f vary in space and may jump across Γ .

- ① Finite Element: Body-fitted mesh (Babuska 1970), modified basis functions (Li 1998)
- ② Finite Difference: Immersed Interface (Leveque 1994), Ghost Fluid (Liu 1999)
- ③ Finite Volume: Cut-cell (Crockett 2011), hybrid (Bochkov 2020)

Underlying logic for finite difference and finite volume type schemes is to **approximate the solution with a Taylor series that obeys interface jump conditions.**

- ① Generalize truncation error analysis of the Taylor series approach to achieve high order accuracy
- ② Simple algorithm for generating geometric information
- ③ Application of the method to a nonlinear elliptic equation from ice sheet dynamics

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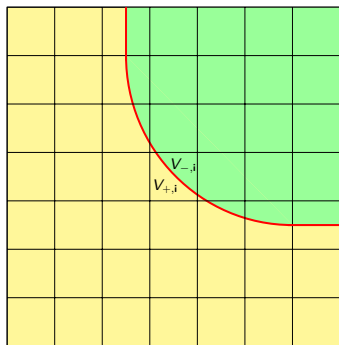
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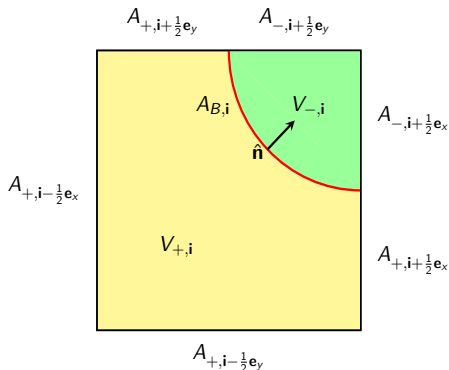
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Finite Volume, Cut-Cell Geometric Description

Γ divides Cartesian cell into two volumes. Solution is represented as **cell averages** over full and partial volumes.



(a)



(b)

Finite Volume Discretization

In each full and partial volume we must satisfy:

$$\langle \alpha u \rangle_{p,i} - \langle \nabla \cdot \beta \nabla u \rangle_{p,i} = \langle f \rangle_{p,i} \quad (4)$$

$$[u]_i = w_i \quad (5)$$

$$[\beta \partial_{\mathbf{n}} u]_i = v_i . \quad (6)$$

where $\langle \cdot \rangle$ indicates an average over a cell volume. Apply Green's theorem to the flux divergence term:

$$\int_V \nabla \cdot \beta \nabla u \, dV = \left[\sum_{\pm, d} (\pm 1) \int_{A_{\pm \frac{1}{2} \mathbf{e}_d}} \beta \frac{\partial u}{\partial x_d} \, dA \right] + \int_{A_B} \beta \nabla u \cdot \hat{\mathbf{n}} \, dA . \quad (7)$$

How we do construct high order stencils for this operator in the finite volume context?

Local Truncation Error Analysis

Suppose we want to construct a stencil for

$$G(\beta, u) = \int_A \beta \nabla u \cdot \hat{\mathbf{n}} \, dA \quad (8)$$

The local truncation error is defined as:

$$\tau = \mathbf{d}_\beta^T \mathbf{S} \mathbf{d}_u - G(\beta, u) \quad (9)$$

where $\mathbf{d}_\beta, \mathbf{d}_u$ are vectors of “data” about β, u : point values, cell averages, boundary and jump conditions. The matrix $\mathbf{S}$ represents a bilinear stencil: we multiply $\mathbf{S}$ on the left by known values of β and on the right by potentially unknown values of u .

Local Truncation Error Analysis

Express u, β as Taylor series expansions:

$$u(\mathbf{x}) = \sum_{|\mathbf{q}| \leq P} c_u^{\mathbf{q}} \mathbf{x}^{\mathbf{q}} + O(h^{P+1}) \quad (10)$$

Multi-index notation:

$$\mathbf{q} = (q_x, q_y), |\mathbf{q}| = q_x + q_y, \mathbf{x}^{\mathbf{q}} = x^{q_x} y^{q_y}, c_u^{\mathbf{q}} = \frac{1}{q_x! q_y!} \frac{\partial^{\mathbf{q}}}{\partial x^{q_x} \partial y^{q_y}} u(\mathbf{x}_0).$$

Compile the data vector $\mathbf{d}$:

$$\mathbf{M}_u \mathbf{c}_u = \mathbf{d}_u + O(h^{P+1}) \quad (11)$$

where $\mathbf{M}$ represents the action on the Taylor coefficients $\mathbf{c}_u$ that matches each piece of data.

Local Truncation Error Analysis

Insert Taylor expansions into integral:

$$G(\beta, u) = \int_A \left(\sum_{|\mathbf{r}| \leq P} c_{\beta}^{\mathbf{r}} \mathbf{x}^{\mathbf{r}} \right) \left(\sum_{|\mathbf{q}| \leq P} c_u^{\mathbf{q}} \left[\frac{\partial \mathbf{x}^{\mathbf{q}}}{\partial x} \hat{n}_x + \frac{\partial \mathbf{x}^{\mathbf{q}}}{\partial y} \hat{n}_y \right] \right) + O(h^P) dA \quad (12)$$

$$= \mathbf{c}_{\beta}^T \mathbf{G}_A \mathbf{c}_u + O(h^{P+1}) \quad (13)$$

where

- ① $\mathbf{c}_u, \mathbf{c}_{\beta}$ are the Taylor expansion coefficients of u, β
- ② $\mathbf{G}_A$ is a matrix that comes from integrating monomial terms

Local Truncation Error Analysis

How do we obtain the matrix $\mathbf{S}$? Insert $\mathbf{M}_u \mathbf{c}_u = \mathbf{d}_u$ into LTE expression:

$$\tau = (\mathbf{c}_\beta^T \mathbf{M}_\beta^T) \mathbf{S} (\mathbf{M}_u \mathbf{c}_u) - \mathbf{c}_\beta^T \mathbf{G}_A \mathbf{c}_u + O(h^{P+1}) \quad (14)$$

to ensure that this error does not exceed $O(h^{P+1})$, we must satisfy:

$$(\mathbf{c}_\beta^T \mathbf{M}_\beta^T) \mathbf{S} (\mathbf{M}_u \mathbf{c}_u) = \mathbf{c}_\beta^T \mathbf{G}_A \mathbf{c}_u \implies \quad (15)$$

$$\mathbf{S} = (\mathbf{M}_\beta^T)^\dagger \mathbf{G}_A \mathbf{M}_u^\dagger \quad (16)$$

because this must hold for any c_u, c_β . The same methodology produces stencils for the rest of the operator.

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Algorithm Implementation

Given order of accuracy P ,

- 1 Generate geometric information
- 2 Generate stencils
- 3 Assemble and solve system

Stencil Construction

The upshot of the analysis is that constructing stencils involves two steps:

- 1 Assembling interpolation matrices $\mathbf{M}_u, \mathbf{M}_\beta, \mathbf{M}_\alpha$
- 2 Assembling matrices $\mathbf{G}$ that result from applying local operator to polynomials

These matrices contain integrals or evaluations of monomials over or at portions of the geometry: cells, faces, centroids. In this context geometry “generation” means calculating and storing these quantities.

The interface is defined as the zero level set of an implicit function $\psi(\mathbf{x})$.
etc romberg

Interface Conditions

Finally: how do we incorporate jump conditions?

- 1 Choose neighborhoods $\mathcal{N}_+, \mathcal{N}_-$
- 2 Let $\mathbf{d}_u$ contain cell averages of u in the neighborhoods, interface jump conditions for cells in neighborhood
- 3 Solve for coupled of Taylor coefficients $\mathbf{c}_{u,+}, \mathbf{c}_{u,-}$:

$$\begin{bmatrix} \mathbf{M}_{u,+} & 0 \\ 0 & \mathbf{M}_{u,-} \\ \mathbf{M}_{J,+} & \mathbf{M}_{J,-} \end{bmatrix} \begin{bmatrix} \mathbf{c}_{u,+} \\ \mathbf{c}_{u,-} \end{bmatrix} = \begin{bmatrix} \mathbf{d}_{u,+} \\ \mathbf{d}_{u,-} \\ \mathbf{d}_J \end{bmatrix}$$

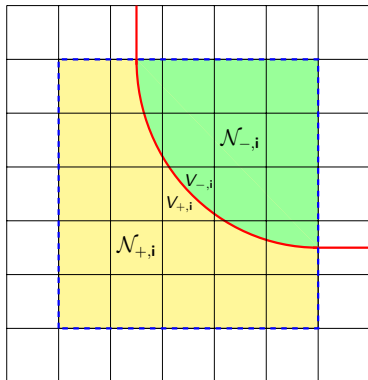


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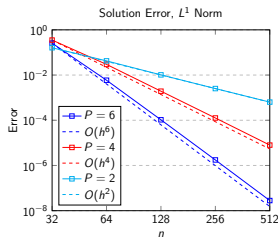
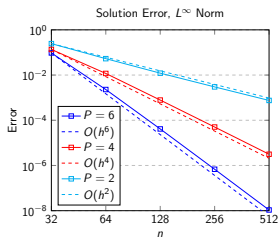
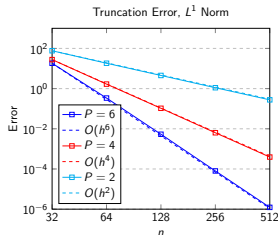
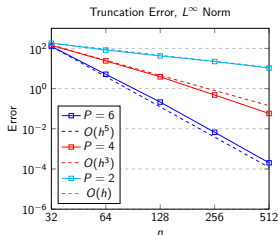
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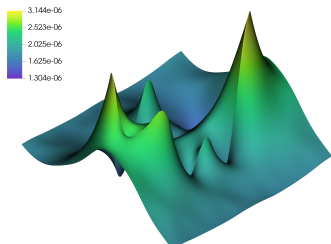
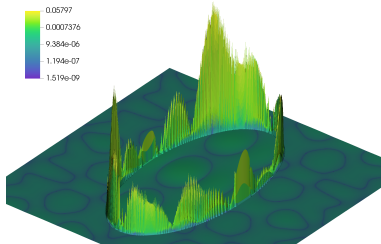
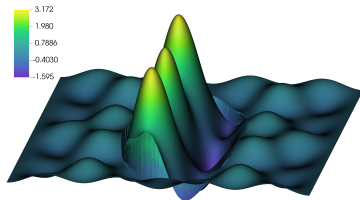
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Truncation and Solution Error Tests



Truncation and Solution Error Tests



Shallow-Shelf Equations

The “grounding line” in ice sheet dynamics is the interface where ice transitions from sitting on bedrock to floating in the ocean. This creates discontinuous behavior at the grounding line, which is a source of error in many numerical ice sheet models.

One approximation to the stress balance equation for ice sheets is the “shallow shelf” equations (cite):

$$-\beta \mathbf{u} + \nabla \cdot (\mu H \mathbf{F}(\mathbf{u})) = \rho g H \nabla s \quad (17)$$

where μ is the nonlinear viscosity, β is the nonlinear friction coefficient, H is the ice thickness, s is the surface of the ice, and $\mathbf{u}$ is the two-dimensional velocity field at the base of the ice.

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